

SIMATS ENGINEERING
ASSESSMENT TOOL 1

(31)

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A
SHORT ANSWER TEST

Description about the assessment tool:

Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

Code of Conduct:

I, K. Sandhiya / 192521378 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

K. Sandhiya
Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	2	2	2	2
Organization	15%	1	1	1	1
Accuracy	10%	1		1	
Clarity	5%				

Total Marks 80 /40

[Signature]
Signature of the Faculty

NAME : K. Sandhya

REG NO : 190521318

ASSIGNMENT 1 COURSE : Computer Networks

COURSE CODE : CSAD7

DATE : 21-04-2026

Q. A transport layer protocol uses a sliding window size of 6 frames.

Given:

Sliding window size = 6 frames

Each frame = 1024 bytes

Total frames = 48

Calculation:

Number of transmissions windows =

Total frames ÷ window size

$$= 48 \div 6$$

$$= 8 \text{ windows}$$

If one frame is lost in every window:

* Retransmission = 1×8

$$= 8 \text{ frames}$$

Answer

* Number of transmission window = 8

* Total retransmission = 8 frames

Explanation

The transport layer is responsible for reliable communication between the sender and receiver. It divides the data into frames or segments and ensures that every frame reaches the destination correctly. The sliding window protocol allows the sender to transmit multiple resulting in a total of 8 retransmissions.

A 12,000-byte IP Packet is transmitted through a router.

Given:

* IP Packet size = 12,000 bytes

* MTU = 1500 bytes

* IP header = 20 bytes

Calculation: Maximum data in each fragment

$$= 1500 - 20$$

$$= 1480 \text{ bytes}$$

Number of fragments:

$$12000 \div 1480$$

$$= 8.1$$

Therefore = 9 fragments

Header overhead:

$$= 9 \times 20$$

$$= 180 \text{ bytes}$$

Answer: No. of fragments = 9

Total IP header overhead = 180 byte

A Transport layer protocol uses a sliding window size of 6 segments.

Given:

Sliding Window size = 6 frames

Total frames = 48

One frame lost in each window

Calculation: Transmission Windows:

$$48 \div 6 = 8 \text{ Windows.}$$

Retransmissions:

$$1 \times 8 =$$

8 frames.

Answer:

No. of transmission windows = 8

Total retransmission = 8 frames.

A multimedia application transfers
a 600 MB Video file.

Given Video file size = 600 MB

- * Video file size = 600 MB
- * Synchronization Checkpoint every 60 MB.
- * Failure occurs after 390 MB

Calculation:

60 MB
120 MB
180 MB
240 MB
300 MB
360 MB

Total checkpoint before failure = 6

Data to ~~be~~ transmit:
 $= 390 - 360$
 $= 30 \text{ MB.}$

Answer:

* Number of checkpoints before failure
6

data transmitted =